

TACNA, Peru

WMO: 847820

Lat: 18.053S

Lon: 70.276W

Elev: 469

StdP: 95.82

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	10.2	7.8	7.0	11.3	8.2	7.1	11.5	6.4	17.6	5.8	17.4	1.2	30	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.5	28.8	20.6	27.9	20.1	27.0	19.6	21.1	27.4	20.4	26.5	19.9	25.9	5.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.6	24.6	18.1	13.8	24.0	17.6	13.3	23.6	63.3	27.5	61.0	26.8	59.1	26.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	6.2	5.7	DB	8.5	30.4	1.2	0.9	7.7	31.1	7.0	31.6	6.3	32.1	5.4	32.8	(4)
(5)			WB	7.5	21.9	1.2	1.1	6.7	22.7	5.9	23.4	5.2	24.0	4.4	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	22.6	23.3	22.3	20.2	18.0	16.1	15.0	15.3	16.2	17.6	19.2	20.9	(6)	
	DBStd	3.09	1.17	1.18	1.18	1.31	1.47	1.28	1.22	1.45	1.34	1.17	0.98	0.98	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	390	0	0	0	2	24	68	102	97	66	28	3	0	(9)	
	CDD10.0	3237	392	374	381	306	247	182	156	163	186	237	277	337	(10)	
	CDD18.3	585	133	141	123	58	13	1	0	1	1	6	31	78	(11)	
	CDH23.3	2227	576	689	499	148	21	2	1	6	2	11	53	219	(12)	
	CDH26.7	254	71	117	52	4	1	1	1	1	0	1	0	6	(13)	
(14) Wind	WSAvg	2.2	2.5	2.5	2.3	2.1	1.9	1.9	1.9	2.0	2.2	2.5	2.6	2.5	(14)	
(15) Precipitation	PrecAvg	11	2	0	1	0	0	1	1	2	2	0	1	0	(15)	
	PrecMax	41	16	4	2	4	6	4	11	14	14	10	3	7	(16)	
	PrecMin	1	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	10	3	1	1	1	1	1	2	3	4	2	1	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	30.1	29.1	27.2	25.2	22.8	21.2	22.9	23.2	24.6	25.8	27.2	(19)	
		MCWB	21.3	21.4	21.1	19.6	18.3	16.6	15.6	17.1	16.8	17.5	18.1	19.4	(20)	
	2%	DB	28.2	29.0	28.0	26.0	23.8	21.1	20.0	20.8	21.9	23.1	24.8	26.2	(21)	
		MCWB	20.1	20.7	20.3	19.0	17.8	15.8	15.2	15.3	15.9	16.7	17.5	18.8	(22)	
	5%	DB	27.2	28.1	27.1	25.0	22.2	20.1	18.9	19.5	20.8	22.1	23.9	25.2	(23)	
		MCWB	19.6	20.1	19.7	18.4	16.8	15.4	14.5	14.8	15.4	16.1	17.2	18.2	(24)	
	10%	DB	26.3	27.1	26.1	24.0	21.2	19.0	17.9	18.2	19.6	21.1	23.0	24.8	(25)	
		MCWB	19.1	19.7	19.2	18.0	16.3	14.8	13.9	14.0	14.9	15.5	16.8	17.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	22.1	21.6	20.1	18.7	16.8	16.2	17.3	17.2	17.7	18.4	20.0	(27)	
		MCDB	28.3	29.0	28.1	25.9	24.3	21.7	20.3	22.2	21.9	23.6	24.8	26.1	(28)	
	2%	WB	20.9	21.2	20.8	19.3	17.8	16.1	15.2	15.7	16.3	17.0	17.8	19.2	(29)	
		MCDB	26.8	27.7	27.0	25.2	23.0	20.6	18.9	19.5	20.8	22.5	23.9	25.4	(30)	
	5%	WB	20.1	20.6	20.1	18.6	17.2	15.4	14.7	14.9	15.6	16.3	17.2	18.6	(31)	
		MCDB	26.0	26.8	26.1	24.5	21.8	19.4	18.0	18.9	19.8	21.5	23.2	24.7	(32)	
	10%	WB	19.5	20.0	19.5	18.0	16.5	14.9	14.1	14.2	15.0	15.7	16.8	18.1	(33)	
		MCDB	25.4	26.0	25.3	23.7	20.6	18.3	17.1	17.9	19.0	20.6	22.4	23.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.5	9.6	9.2	8.3	7.5	7.4	7.9	8.3	8.9	9.2	9.4	(35)	
		MCDBR	9.8	9.8	9.9	9.8	9.0	8.4	8.2	8.9	9.5	9.5	9.7	9.6	(36)	
	5% WB	MCWBR	4.4	4.2	4.5	4.7	4.4	4.2	4.2	4.5	4.6	4.4	4.4	4.4	(37)	
		MCWBR	9.4	9.4	9.6	9.5	8.8	8.1	7.8	8.7	9.0	9.3	9.4	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.504	0.507	0.491	0.435	0.373	0.354	0.354	0.379	0.401	0.426	0.428	0.477	(40)	
		taud	2.053	2.052	2.093	2.208	2.326	2.385	2.376	2.279	2.217	2.150	2.175	2.080	(41)	
	Ebn at Noon	Ebn at Noon	852	839	829	838	859	859	871	879	896	900	912	874	(42)	
		Edh at Noon	181	179	166	138	114	104	107	127	144	160	159	176	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	7.24	6.85	6.75	5.95	5.04	4.34	4.39	5.21	6.18	7.02	7.82	7.75	(44)	
	RadStd	RadStd	0.53	0.55	0.41	0.40	0.23	0.25	0.27	0.28	0.34	0.27	0.32	0.37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page